

WELLSLING · REQUIREMENTS

WellSling requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsling		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 3 status for every requirement from WSL-CAL-001; R2 restated for the kit's own tripod; R9 restated as a value-engineering target; R10 to R13 added (WSL-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

WellSling requirements

Twelve of the thirteen requirements are met on paper or by design. R9 is reported against the value-engineering target, which the constructable design exceeds by USD 1,504, and R8 (operation after a short briefing) can only be shown in a user trial. Results come from WSL-CAL-001; tags in square brackets point to its sections.

Safety: WellSling lifts a heavy, frightened animal over an open well. Nobody goes down the well at any stage. The kit is an open engineering reference, not certified lifting or rescue equipment, and it must be proof-loaded before any use (TRL 4).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	No entry	Bands passed, rigged, lifted and landed entirely from the rim	Trial with a weighted dummy in a test pit	Met by design
R2	Lifting capacity	Design animal 600 kg (648 kg at the hook with rigging), with the hook line leaning up to 10 deg; proof-loaded to 1.5 times	Proof-load test of tripod, winch, rope and rigging	Met on paper [B1, B5, C5, C6]
R3	Band spreads the load	Each band at least 200 mm (8 in) wide; average pressure on the animal stated	Design review with a veterinary partner	Met on paper: two 240 mm bands, about 17 kPa [E6]
R4	Passer reach	Passes the band under an animal up to 15 m (50 ft) below the rim	Trial in a test pit with a dummy	Met on paper: 16.5 m [G3]
R5	Well span	Straddles wells up to 3 m (10 ft) across, every pad and bearer at least 350 mm back from the rim	Fit check on sample wells	Met: pads 700 mm, bearers 400 mm back [C9, F4]
R6	Fast setup	Three people ready to lift within 20 minutes of arrival	Timed drill	Met on paper, at the limit: 20 min estimate [G4]
R7	Controlled lift	Load held by the winch brake at any point when the handle is released	Brake hold test at proof load	Met by selection of a load-pressure brake
R8	Simple operation	Villagers complete a lift after a 15-minute briefing	User trial with first-time operators	Not shown on paper; to verify at TRL 4
R9	Value-engineering target	USD 1,500 for the whole kit (a hypothetical control target, not a limit)	Costed bill of materials	Estimated USD 3,004: over the value-engineering target by USD 1,504 [G2]
R10	Tripod stability	No foot lifts with the hook line leaning 10 deg in any direction	Lean test at proof load	Met on paper: tips only at 19.3 deg [C4]
R11	Hand effort	Crank force at most 250 N throughout the lift	Force gauge on the handle	Met on paper: 196 N at most [B6]
R12	Landing clearance	Hooves at least 150 mm above the deck at the landing height, so the bridge slides under	Fit check with a dummy	Met on paper: 162 mm [F5]
R13	Carry weight	No single piece heavier than 35 kg, so two people carry any piece	Weigh the parts	Met on paper: 31.6 kg [G1]

■ Assumptions

- The design animal is 600 kg, the upper end of a grown cow or buffalo; a heavier animal is outside the rating.
- The ground around the well bears at least 75 kPa under a timber pad; a softer site needs larger pads.
- A plain webbing band with a loose felt sleeve can lift a large animal for the length of a lift without injury when two bands share the load. This is to be confirmed with a veterinary partner.
- Most open wells where animals fall are 3 m across or less.
- A village or cooperative keeps the kit in one known place and practises with it.